

[UNTITLED]

A CSARCH2 Proposal by Group 8 of S40: Dela Cruz, Ramos, Tengco, Vicencio, Willie

TOPIC THEME:

Our group's topic, under the category of 'How it Works', will be: **How SSDs work**. From this topic, the proposed exhibit will highlight the specific advantages of the Solid State Drive (SSD), by contrasting its features with a mechanical Hard Disk Drive. In addition, the proposed exhibit will cover the internal intricacies of an SSD, specifically how data is written and stored within NAND Flash memory, as well as the role of an SSD controller within an SSD's functions.

TECH STACK PLAN:

Proposed Interactive Elements:

Interactive Element	Detailed Description
Multiple-choice quiz	The exhibit will feature a multiple-choice quiz coded with React components (tentative) that will only come into view (i.e: client:visible, placed at the end of the exhibit's page) once the user is finished scrolling/reading through the exhibit itself. The quiz will test the user's knowledge in regards to how SSDs work and will show the user's total understanding of the topic in the form of a final score that will appear after the user has finished taking the quiz. Additionally, we plan to add a feature in the form of a highlighted inline explanation that will explain the user's mistake should they choose the wrong answer.
Video Gallery Comparison	For the exhibit's section on the comparisons between a SSD and an HDD, we plan to incorporate a multimedia gallery. Within this gallery, it will utilize side-by-side video comparisons to highlight the SSD's speed advantages in real-world scenarios as compared to a mechanical HDD. This will include a comparison between the respective read/write times of particularly large file transfers, the loading times of computationally demanding programs, and vice versa. These videos will be mainly sourced by our own group, with benchmark footage from external reviewers being potentially used (with the appropriate copyright citations) if our group is unable to record certain videos.

Tentative Style Guide Snapshot/Proposed Virtual Exhibit Design Layout:

Main Page

 **Solid State Drives** by Dela Cruz, Ramoz, Tengco, Vicencio, Willie

Introduction to SSDs

by Dela Cruz, Ramoz, Tengco, Vicencio, Willie

What are SSDs?

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A 2TB Samsung NVMe SSD

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Quiz Page

✿ Solid State Drives by Dela Cruz, Ramoz, Tengco, Vicencio, Willie

Multiple-choice Quiz

Sample Quiz

Name

First Name

Last Name

Question 1

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- ☐ Choice A
- ☐ Choice B
- ☐ Choice C
- ☐ Choice D

Question 2

Lorem ipsum dolor sit amet

- ☐ Choice A
- ☐ Choice B
- ☐ Choice C
- ☐ Choice D

Submit

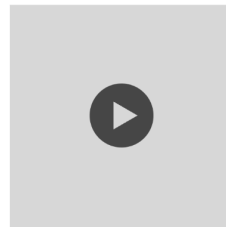
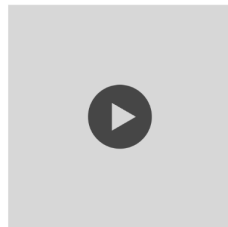
Video Gallery

✿ Solid State Drives by Dela Cruz, Ramoz, Tengco, Vicencio, Willie

HDDs Vs. SSDs

Video Gallery

HDDs Vs. SSDs - Read/Write Times



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Mobile Responsive Layout (optional?):

Main Page

 **Solid State Drives** by Dela Cruz,
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Introduction to SSDs

What are SSDs?



A 2TB Samsung NVMe SSD

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Quiz Page



Solid State Drives by Dela Cruz,
Ramos, Tengco, Vicencio, Willie

Multiple-choice Quiz

Sample Quiz

Name

First Name

Last Name

Question 1

Lorem ipsum dolor sit amet

☐ Choice A

☐ Choice B

☐ Choice C

☐ Choice D

Question 2

Lorem ipsum dolor sit amet

☐ Choice A

☐ Choice B

☐ Choice C

☐ Choice D

submit

Video Gallery Page

**Solid State Drives** by Dela Cruz,
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HDDs Vs. SSDs

HDDs Vs. SSDs - Read/Write Times



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